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Forum response—commentary

The case against renewed ocean iron fertilization field trials

Dialogues on Climate Change
1–8
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DOI: 10.1177/29768659261498319
journals.sagepub.com/home/dcc

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Abstract

Buesseler et al. call for larger and longer ocean iron fertilization (OIF) trials to assess climate-relevant carbon dioxide removal potential. We argue that such trials may document local ecosystem responses but cannot resolve the two central uncertainties for climate policy: removal potential and downstream consequences. Net atmospheric CO_2 removal depends on carbon export, air–sea re-equilibration, circulation, downstream nutrient redistribution, and ecosystem responses occurring beyond trial boundaries. Global models representing these processes consistently find removal potential to be low, and field trials cannot overturn that evidence. The same models identify delayed and far-field impacts on oxygen and ecosystems under basin-scale OIF, which trials can neither evaluate nor prevent, deferring judgment to the models they are intended to improve. Progressively larger OIF trials are therefore unwarranted. Research should instead prioritize observations of naturally fertilized systems and better models, which could test the controls on OIF carbon yield without deliberate perturbation.

Keywords

Ocean iron fertilization, marine carbon dioxide removal, ocean carbon cycle

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Introduction

Carbon dioxide removal (CDR) will be necessary to meet ambitious climate targets (IPCC, 2022: Chapter 12). Ocean iron fertilization (OIF) stimulates CO_2 fixation in iron-limited surface waters, lowering the partial pressure of CO_2 at the sea surface (pCO_2) and promoting uptake of atmospheric CO_2, while a fraction of the fixed carbon sinks to depth, where it is temporarily stored (Buesseler et al., 2026). Most assessments place its upper-end potential at 0.1–1 Gt CO_2 year^{-1}, under idealized scenarios assuming near-complete relief of iron limitation, with a few reaching several gigatonnes (IPCC, 2022; NASEM, 2022), that is, at most a few percent of current anthropogenic CO_2 emissions (~42 Gt CO_2 year^{-1}; Friedlingstein et al., 2025). OIF is nonetheless still being advanced as a CDR option.

Buesseler et al. (2026) argue for a new generation of OIF field trials, larger (>1000 km^2) and longer (3–6 months) than any to date (25–300 km^2 and 10–40 days; Boyd et al., 2007), to quantify both the efficacy of iron fertilization in delivering durable, additional carbon removal and its impacts on marine ecosystems. OIF is already the subject of ethical and governance debates that must be addressed (Boettcher et al., 2021; Cooley et al., 2023; Loomis et al., 2022); here we question the technical feasibility and scientific value of the trials themselves, considerations that are central to the international frameworks designed to assess whether such research is scientifically justified (e.g. London Protocol; IMO, 2010). Buesseler et al. (2026) identify several major uncertainties to be resolved before OIF could advance toward proof of concept. Yet 30 years of ocean biogeochemistry studies (Boyd et al., 2007; Cullen and Boyd, 2008; de Baar et al., 2005; Martin, 1990; Strong et al., 2009; Yoon et al., 2018) suggest two conclusions. First, the structural uncertainties surrounding OIF's removal potential cannot be resolved by these trials. Second, the trials can neither evaluate the impacts of OIF at climate-relevant scales (i.e. global and continuous) nor mitigate the harmful effects they might cause. A single thread runs through both: the trials would defer their central questions to global biogeochemical models, which have repeatedly shown this potential to be low even under infeasibly favorable assumptions (Aumont and Bopp, 2006; Tagliabue et al., 2023; Zahariev et al., 2008).

Field trials cannot constrain the central uncertainties of OIF's removal potential

To constrain the additionality and durability of carbon export, a trial must not merely detect export and its depth, but isolate the fraction caused by fertilization. Yet only a small share of the additional biomass produced following the iron addition is likely to sink: blooms may be limited by other nutrients, such as silicic acid, and much of the biomass is consumed by grazers first (Boyd et al., 2004; Martin et al., 2013). Even when carbon does sink, little reaches greater depths: in the region the authors highlight, only 20% to 50% of the particle flux at 150 m arrives at 500 m, with large differences between sites (Buesseler et al., 2007). The authors themselves acknowledge that the expected increase at 500 m may fall "within the range of natural POC flux variability," making the fertilization signal difficult to distinguish from the background. More fundamentally, increased export is not guaranteed: among 13 historical OIF trials, only EIFEX (the 2004 European Iron Fertilization Experiment) produced an unambiguous increase in deep carbon export (Yoon et al., 2018).

Ocean physics adds to the challenge. A fertilized patch is not a sealed test tube; turbulent stirring stretches it into thin filaments over hundreds of kilometers (Lévy et al., 2018), more strongly at the surface than at depth, spatially decoupling the surface pCO_2 drawdown from any deep export. During SOIREE, the patch stretched from about 7 km to more than 150 km across within six weeks (Abraham et al., 2000), and within six months a patch could spread to 100–400 times its initial area (Groeskamp et al., 2020), drifting out of any realistic instrument array. The same stirring mixes fertilized and surrounding waters, blurring the line between perturbed and control regions (Boyd et al.,

2023), while finescale vertical fluxes may return part of the exported material to the surface (Resplandy et al., 2019). Disentangling this signal from natural variability, seasonal ventilation, ongoing climate change, and extreme events would be extremely difficult (Fennel, 2026; Tagliabue et al., 2023). These limitations weaken the scientific rationale for new field trials at the scales now being proposed.

The uncertainties that govern OIF's net climate benefit arise from processes spanning local to basin scales and days to years, an entanglement beyond the reach of trials that are local and short by nature (Gnanadesikan and Marinov, 2008). The authors propose to concentrate the trials on export depth, on the grounds that it governs storage durability. But how long exported carbon stays out of the atmosphere depends also on circulation and on where it is remineralized (Siegel et al., 2021): in the Southern Ocean, two-thirds of the carbon exported to 1000 m can be re-exposed within a century (Robinson et al., 2014). Export depth is therefore useful but insufficient to quantify what the authors call a "centennial tonne": carbon additionally exported and kept from the atmosphere for at least 100 years. That will require relatively high-resolution global models, whose representations of circulation differ substantially and remain a major source of uncertainty in the biological carbon cycle.

Measuring export is not measuring CO_2 removal, still less net climate benefit. Export lowers sea surface pCO_2 locally, but atmospheric uptake follows over months to years and may remain incomplete if fertilized waters are subducted before they re-equilibrate (Bach et al., 2023; Zhou et al., 2025); the authors acknowledge that this air–sea flux cannot be assessed from local observation alone. A "centennial tonne" of export is therefore not a "centennial tonne" of removal. The benefit also depends on far-field effects a local trial can neither observe nor confidently attribute: fertilization raises productivity inside the patch while depleting nutrients that would have supported it downstream (Jin et al., 2008; Noh et al., 2025; Tagliabue et al., 2023), and may alter the production of greenhouse gases such as N_2O, partially offsetting the expected radiative benefit (Law, 2008). The uncertainties that determine OIF's real climate effectiveness therefore operate precisely at the scales field trials cannot capture.

Such experiments would not necessarily solve the modeling problem they are intended to address. The authors present new trials as a way to improve ocean biogeochemical models, arguing that they poorly represent many processes controlling the biological carbon cycle (Henson et al., 2022). But improving models' process representations requires observations that isolate specific processes and constrain them one by one, from particle aggregation to grazing, remineralization, and air–sea exchange (Siegel et al., 2016). A fertilization trial does the opposite: it would impose a perturbation on an already variable and incompletely characterized ecosystem, producing an integrated response in which these processes are entangled. Such a response may show what happened in one place at one time, but it would not reveal which process is misrepresented in the model or how its representation should be corrected.

The authors see "considerable merit in learning by doing," but learning requires that the experiment isolate what is to be learned. Here the quantity of interest is precisely the one the trial cannot separate from natural variability, cannot follow to its centennial horizon, and cannot distinguish from an unmeasured and highly dynamic far field. In this design, "doing" does not resolve the uncertainty. It produces an aggregate signal whose large-scale interpretation must still be handed back to the global biogeochemical models that the trials were supposed to improve.

More importantly, the limitations of current models do not make the broad assessment of OIF's limited removal potential in the peer-reviewed literature unreliable. Although global ocean biogeochemical models necessarily simplify many processes, they play a central role in assessing the ocean carbon cycle (Friedlingstein et al., 2025) and arguably represent the key controls on OIF carbon removal well enough to estimate its order of magnitude. Across a range of model structures and idealized fertilization scenarios, they consistently find limited removal potential, even under unrealistically optimistic assumptions such as sustained relief of iron limitation or large-scale Southern Ocean fertilization (Aumont and Bopp, 2006; Oschlies et al.,

2010, 2025; Tagliabue et al., 2023; Yu et al., 2026; Zahariev et al., 2008). This consensus is corroborated by the paleoceanographic record: even millennia of basin-scale natural iron fertilization by dust during glacial periods produced only a small additional atmosphere-to-ocean carbon flux (~0.01 Gt C $year^{-1}$; Boyd, 2008).

The trials can neither evaluate nor mitigate OIF's impacts

The trials are also presented as a way to determine, through a "go/no-go" framework, whether OIF can cause long-term alterations to ocean ecosystems. They may fail on two counts: they cannot assess the impacts that matter most at climate-relevant scales, and the framework cannot mitigate the harm they might cause.

The impacts that determine whether OIF is safe for climate-relevant CDR operate at scales a field trial cannot access. A trial is local and lasts months, whereas the consequential impacts are far-field and unfold over years to decades. Fertilization depletes nutrients that would otherwise sustain productivity downstream, with consequences for fisheries and higher trophic levels (Gnanadesikan and Marinov, 2008; Tagliabue et al., 2023); these responses are also intermittent and patchy, making them especially hard to detect and attribute. Large-scale OIF could further intensify deoxygenation and acidification beneath fertilized regions and in distant waters connected by circulation (Oschlies et al., 2010, 2025), acting in the same direction as climate change and so amplifying existing stressors (Tagliabue et al., 2023). None of this can be observed within a single patch over a few months, as Buesseler et al. themselves acknowledge.

Crucially, these are not speculative risks: they are consistently identified across the same global models the authors rely on elsewhere, and they scale with the intervention. Effects that appear negligible in a small patch can reach, for example, a ~10% loss of Southern Ocean oxygen at climate-relevant scale (Oschlies et al., 2010, 2025). A trial never measures this scale-to-impact relationship, cannot demonstrate that large-scale deployment is safe, and, if perceived as successful, could provide false reassurance. The assumption that the system simply "resets" after iron addition stops is equally untestable: recovery of slow processes such as subsurface deoxygenation or ecosystem reorganization cannot be observed on a trial's timescale and is postulated rather than shown; climate-relevant OIF would in any case entail continuous fertilization with no resets. We recognize that scaling experiments progressively is standard practice, and that every CDR option involves trade-offs policymakers must weigh. But the case for larger trials depends on their resolving the uncertainty that justifies them, and here they cannot: the decisive impacts lie beyond the scope of the proposed trials, so their diagnosis would again be delegated to the very models whose limitations the authors elsewhere invoke. Those model limitations should not be used to dismiss ecosystem risks that are difficult to observe and quantify but remain highly plausible.

Even for the impacts a trial can detect, the proposed off-ramp is reactive rather than preventive. Monitoring toxins, oxygen, pH, or food-web shifts may help decide when to stop an experiment, but cannot reverse changes already set in motion, and some of the relevant harms are local and observable yet potentially irreversible. Iron enrichment stimulated domoic acid production by *Pseudo-nitzschia* in seawater from Ocean Station Papa (Trick et al., 2010), and concentrations reached ecologically concerning levels during SOFeX (Silver et al., 2010); iron-driven food-web shifts and subsurface deoxygenation could likewise persist after iron addition stops. A go/no-go rule can therefore limit further exposure, but cannot make the intervention safe once an impact has been triggered, let alone remediate far-field consequences the trial never observes. That limitation grows with sustained, climate-relevant deployment.

Conclusion

Taken together, these limitations make the scientific case against a new generation of OIF field trials. Such experiments may document local ecosystem responses, but they cannot resolve the uncertainties that determine whether climate-relevant OIF could deliver durable, additional, net carbon removal, nor demonstrate its environmental safety at that scale. Instead, their interpretation would again depend on global biogeochemical and ecosystem models whose limitations are invoked to justify the trials in the first place. This creates

an unfavorable balance between scientific return and environmental risk. Uncertainty should not license progressively larger interventions when those interventions cannot answer the questions on which their acceptability depends.

The priority should instead be to improve our understanding of the core mechanisms governing carbon export, transformation, and storage in the ocean. That knowledge is needed regardless of OIF: it is essential to anticipate how the ocean's biological carbon cycle will respond to ongoing and future pressures. It would be better pursued through targeted observations and model development. For example, research should prioritize targeted process studies and sustained observations of naturally fertilized systems, including waters around the Kerguelen and Crozet archipelagos, and the iron-enriched waters surrounding Pacific islands (Bourdin et al., 2026). In parallel, models should be used to test how uncertainties in carbon export, mesoscale physics, and biogeochemical feedbacks modulate the estimated carbon yield of OIF. Such efforts would provide a stronger empirical basis for improving biogeochemical models without relying on progressively larger deliberate perturbation experiments whose interpretation remains structurally ambiguous.

We offer these comments in a constructive spirit and with great respect for the authors of the proposal advanced by Buesseler et al., many of whom are leading scientists whose contributions have substantially advanced our understanding of the ocean carbon cycle. We hope that this critique will contribute to an open and constructive scientific dialogue and help direct future research efforts and investments toward the approaches most likely to yield robust and policy-relevant knowledge. Ultimately, we share the same objective: identifying effective, scientifically sound, and societally responsible pathways for climate mitigation.

Acknowledgments

We thank B. B. Cael for reading the manuscript and for expressing his support for its message.

Funding

The authors disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: M.P. was supported by the French Priority Research Program "Ocean and Climate," funded by France 2030 and operated by the Agence Nationale de la Recherche.

Declaration of conflicting interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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